

~~Alternative statistical and data analyses~~

Network studies applied to microbial communities interaction

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Funding

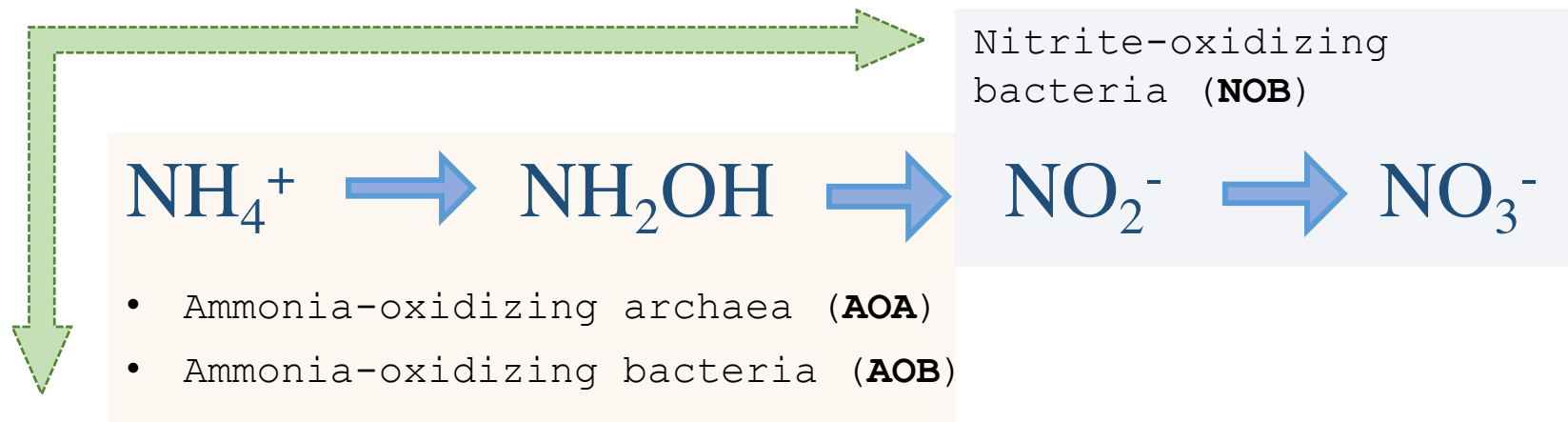
Sponsorship

Support

Why to study interactions?

Modularity of metabolic pathways is prevalent in microbial communities

Nitrification (chemoautotrophs)



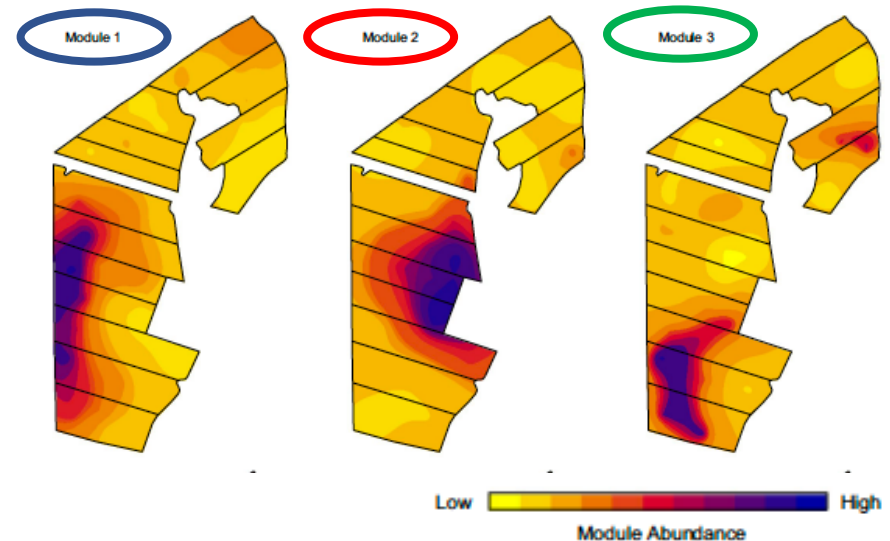
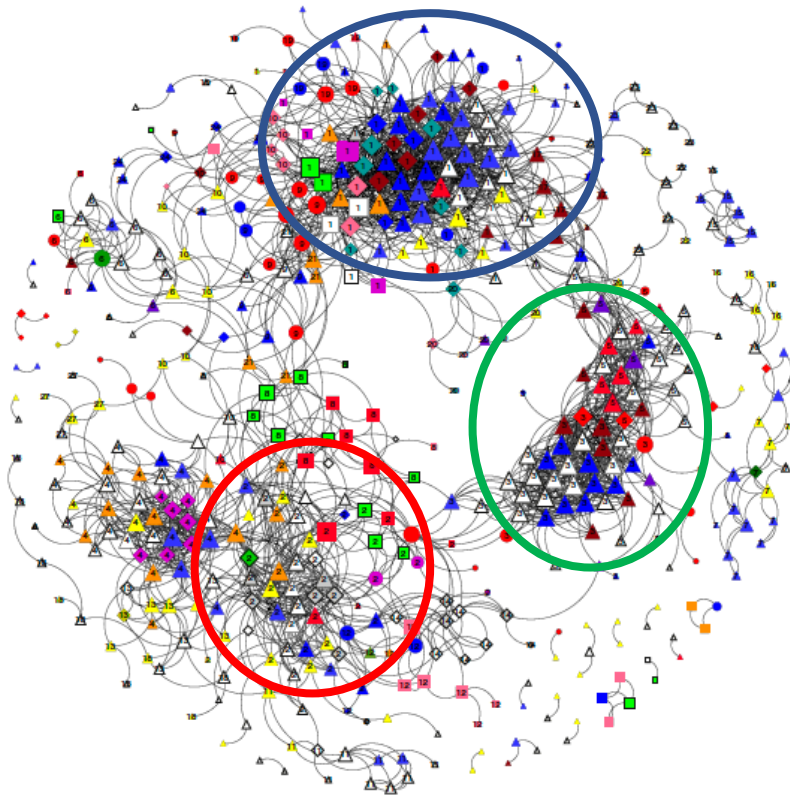
Mutualistic interaction

More complex in some cases (e.g. NOB urease, heterotrophs)

Why to study interactions?

Biological interactions within and between functional groups can play a vital role in microbial community structure.

Network of Pearson correlations



"(...) distinct modules of co occurring AO and NOB groups occupying disparate areas, with each module dominated by different lineages and associated with different edaphic factors."

Why to study interactions?

Biological interactions within and between functional groups can play a vital role in microbial community structure.

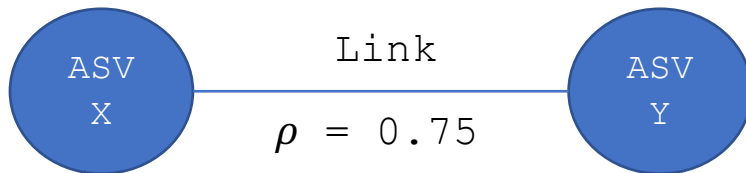
“Although network analysis **does not explain causal mechanisms** (...) these findings provide useful **starting points for future manipulation experiments** that can confirm, or refute, inferences on lineage-specific interactions and mechanisms underlying nonrandom associations within and between functional groups...”

In Jones and Hallin, 2018, Molecular Ecology.

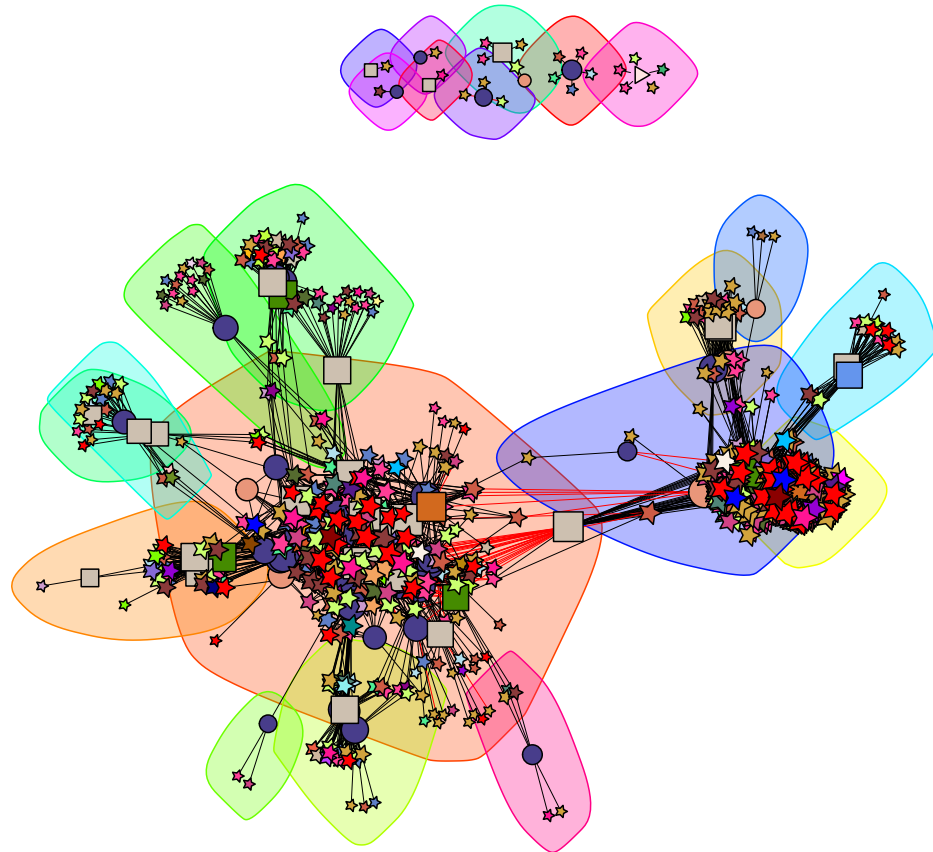
Interaction networks

- **Spatial arrangement of multiple interactions** (as opposed to isolated pairwise interactions). Tightly linked **groups of organisms** will cluster together.
- **Subcommunities**/modules/clusters

Co-occurrence/interaction networks are **usually based on some correlation unit** (e.g. Spearman or Pearson).



Correlations Network Between Nitrifying Genera and the Overall Community

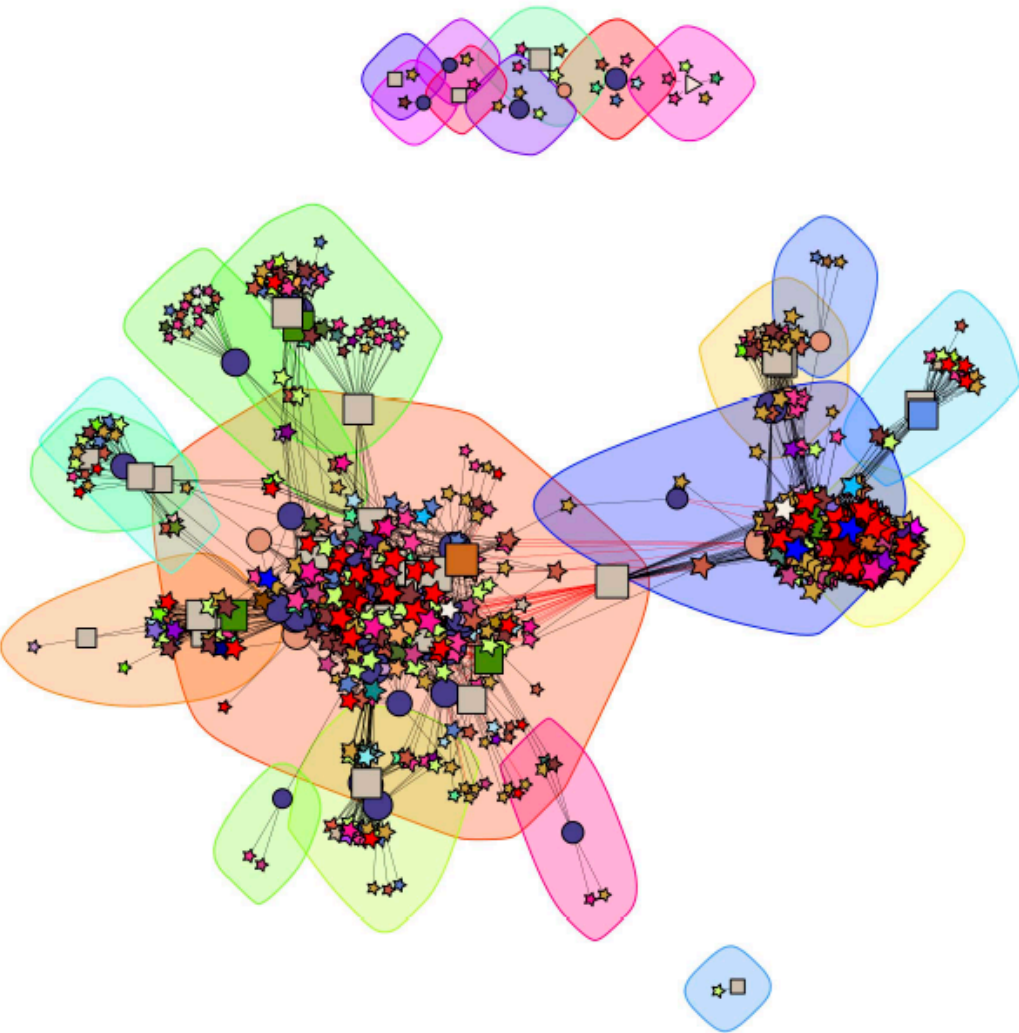


Semedo et al., 2021, Frontiers in Microbiology

Interaction networks

Correlations Network Between Nitrifying Genera and the Overall Community

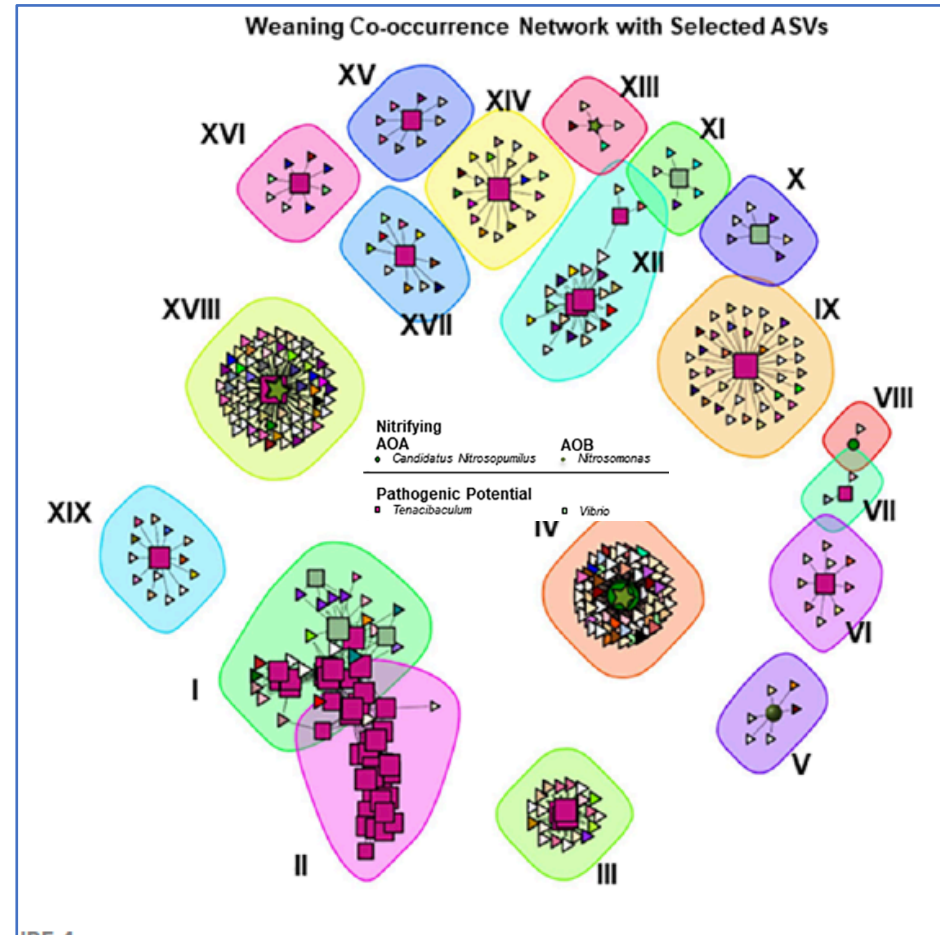
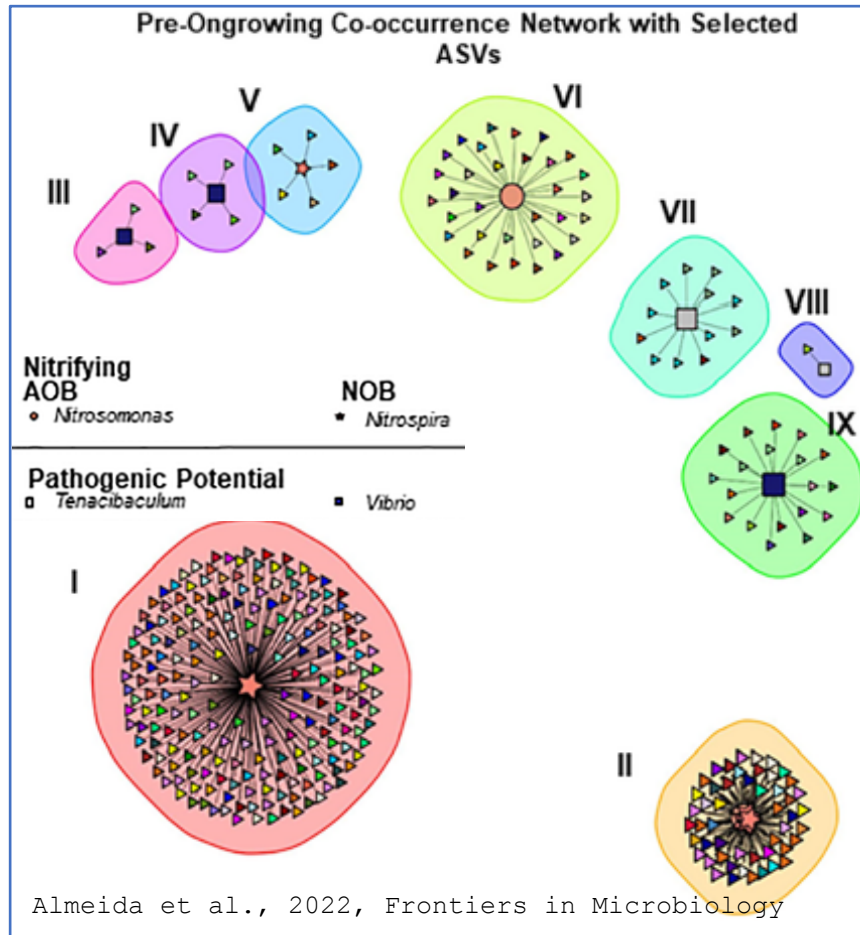
Are unclassified AOA isolated from NOBs?



NITRIFYING GENERA		
AOA (Thaumarchaeota)		NOB
● Nitrosopelagicus		■ Nitrospira (Nitrospirae)
● Nitrosopumilus		■ LS-NOB (Nitrospirae)
● Unclassified		■ Nitrospina (Nitrospirae)
AOB (Gammaproteobacteria)		■ Unclassified (Nitrospirae)
△ FS142-36B-02		
OVERALL COMMUNITY		
Gammaproteobacteria	Planctomycetes	Other Unclassified
★ Endoecksteinascidia	★ JL-ETNP-F28	★ Unclassified Euryarchaeota
★ Enhydrobacter	★ Pir4 lineage	★ Unclassified Gemmatimonadetes
★ Marinobacter	★ Rhodopirellula	★ Unclassified Hydrogenedentes
★ Methylophaga	★ Unclassified	★ Unclassified Kiritimatiellaeota
★ OM43 Clade	★ Urania-1B-19	★ Unclassified Margulisbacteria
★ Pseudohongiella	Verrucomicrobia	
★ SUP05 Cluster	★ Verruc-01	★ Unclassified Marinimicrobia
★ Woestia	★ Cerasicoccus	★ Unclassified PAUC34f
★ Unclassified	★ MB11C04	★ Unclassified Schekmanbacteria
Alphaproteobacteria		★ Unclassified Anck6
★ AT-s3-44	★ Unclassified	★ Unclassified Chlamydiae
★ Clade Ia	Actinobacteria	
★ Clade Ib	★ Actinomarina	★ Unclassified Chloroflexi
★ Magnetospira	★ Mycobacterium	★ Unclassified Crenarchaeota
★ Unclassified	★ Unclassified	★ Unclassified Proteobacteria
Deltaproteobacteria		★ Unclassified Bacteria
★ OM27 Clade	Bacteroidetes	
★ Unclassified	★ Gramella	
Cyanobacteria		★ Marinococcus
★ Prochlorococcus	★ Unclassified	
★ Unclassified	Acidobacteria	
Deinococcus-Thermus		★ PAUC26f
★ Truepera	★ Unclassified	

Interaction networks

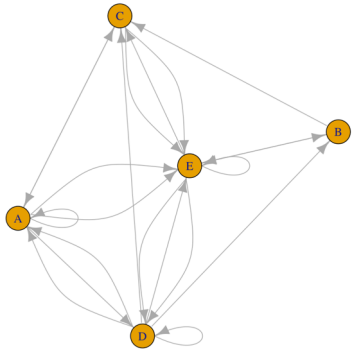
Are two potentially pathogenic genera tightly linked or negatively correlated?



Not negatively correlated (no evidence of exclusion competition) and data suggests positive correlations in one of the systems (weaning)

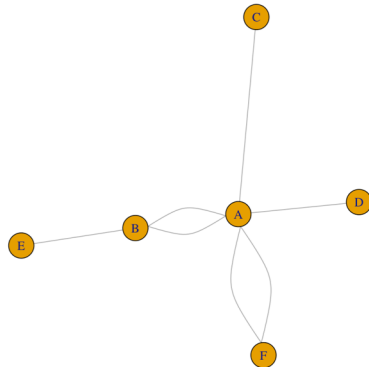
Types of networks

Directed



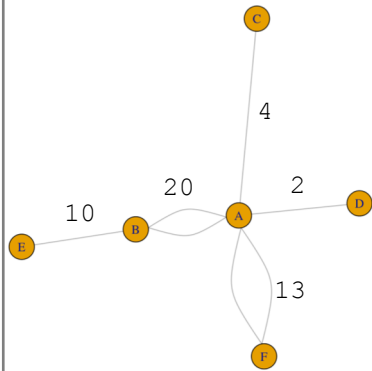
e.g. Twitter

Undirected

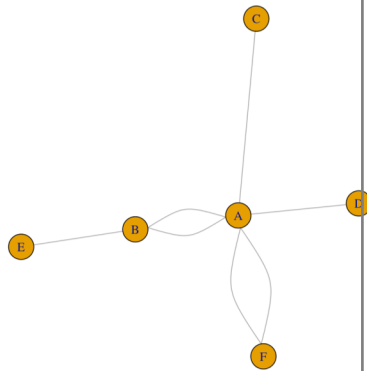


e.g. Facebook

Weighted

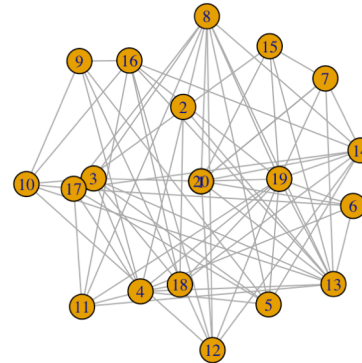


Unweighted

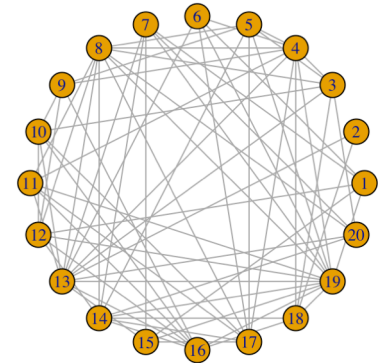


Layout algorithms ("Force-based")

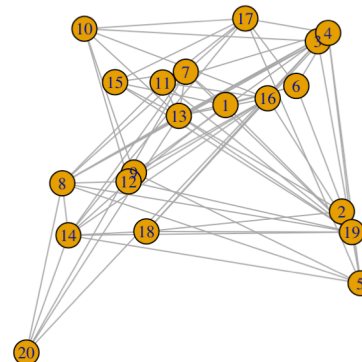
sphere



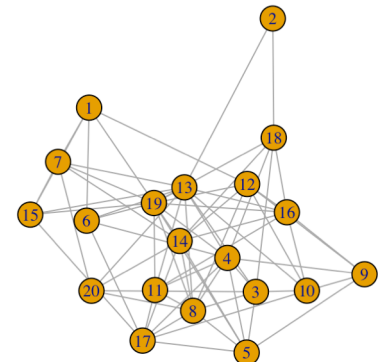
circle



random



fruchterman.reingold



Community detection: Louvain method, Leiden algorithm, etc.

Networks glossary

- **Node (or vertex):** connecting points
- **Link (or edge):** connecting lines
- **Module:** subcommunity/group/cluster of tightly linked nodes
- **Degree:** total number of links (per node, per network, per module)
- **Network layout:** coordinates algorithm used to spatially arrange nodes and links in order to minimize crossing and overlap.
- **Hub nodes or hub taxa:** highly connected nodes
- **Module-hub:** highly connected node within a module ("specialists")
- **Module connector:** node with multiple connections between modules ("generalists")

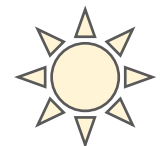
Concerns, limitations, and other approaches

- **Autocorrelation** (unless you want to test it):
Nodes interacting due to a third major force (e.g. habitat effect).
- Correlation coefficient and **threshold value**.
- Do you have a **specific hypothesis or research question**?
Avoid performing a network analysis "just to check".

Other approaches that can evaluate interactions:

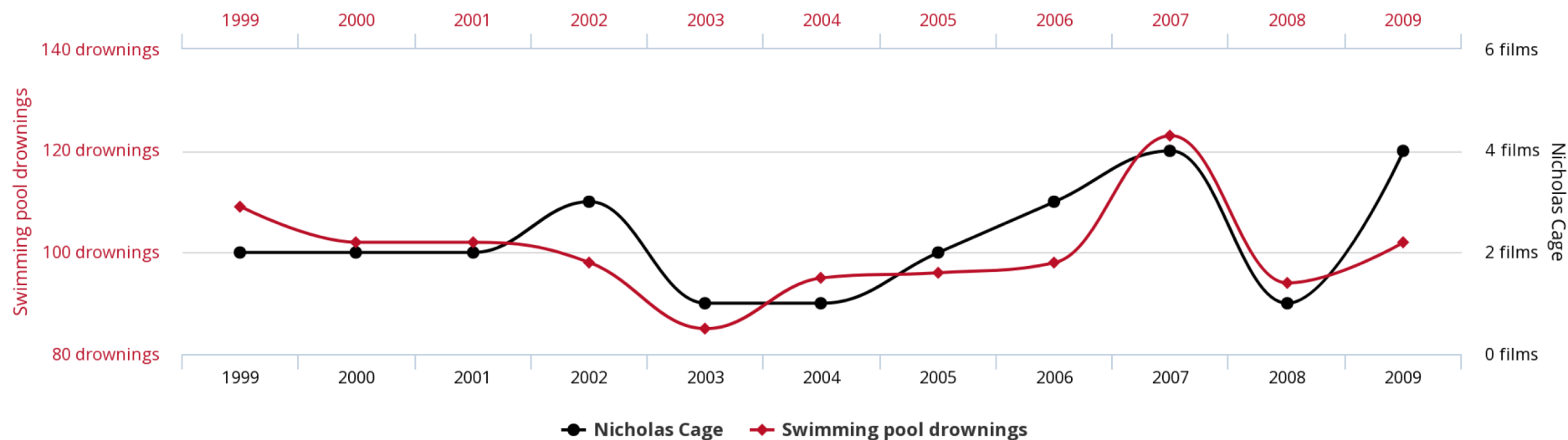
- Structural equation modelling
- Akaike-information criterion (AIC) analysis
- Association rules

Spurious correlations



- **CORRELATION IS NOT CAUSATION !**
- **OCCASIONAL OBSERVATION DOES NOT REPLACE EXPERIMENTATION !**

Number of people who drowned by falling into a pool
correlates with
Films Nicolas Cage appeared in



Data source: US CDC and IMDB

tylervigen.com

Questions?